第4章. 波导 & 光纤.

4.1 (锯) Helmholtz eq: $\nabla^2 E + k^2 E = 0 \rightarrow (\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2})E + (k^2 - k_z)E = 0$.

边界条件 $\nabla \cdot E = 0$; $n \times E = 0$.

分离变量 $u(x,y) = X(x)Y(y)$. $\begin{cases} X'' + k_x^2 X = 0 \\ Y'' + k_y^2 Y = 0 \end{cases}$

$k^2 = k_x^2 + k_y^2 + k_z^2$.

$\therefore$ ...

$\therefore$ $E_x = \sim$ $E_y = \sim$ $E_z = \sim$

$k_x = m\frac{\pi}{a}$ $m = 0, 1, 2, \cdots$ $k_y = n\frac{\pi}{b}$. $n = 0, 1, 2, \cdots$.

半波数目

k_x, k_y ↗ 以致 $k_x^2 + k_y^2 > k^2$. 由 $k^2 = k_x^2 + k_y^2 + k_z^2$, $k_z \rightarrow$ 虚数.

传播因子 $e^{ikz} \rightarrow e^{-kz}$ (衰减因子). 电磁波传播不断衰减。

$\therefore$ 电磁波有频率下限. —— 截止频率.

$$\omega_{c,m,n} = \frac{1}{\sqrt{\varepsilon\mu}}\sqrt{k_x^2 + k_y^2}$$

$$= \frac{\pi}{\sqrt{\varepsilon\mu}}\sqrt{(\frac{m}{a})^2 + (\frac{n}{b})^2}$$

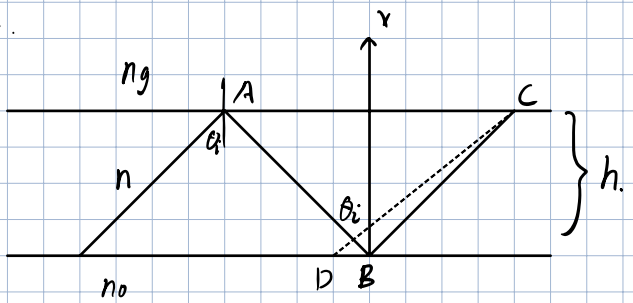
$$\lambda_{c,m,n} = \frac{2\pi}{\sqrt{k_x^2 + k_y^2}} = \sim.$$

4.2 薄膜波导

(1) 全反射条件.

$\theta_i > \theta_c$.

(2) 谐振条件.



相位跃变

谐振条件: $2nhk_0\cos\theta_i + \delta_1 + \delta_2 = 2m\pi$

$\theta_i \longleftrightarrow m$ 一种模式.